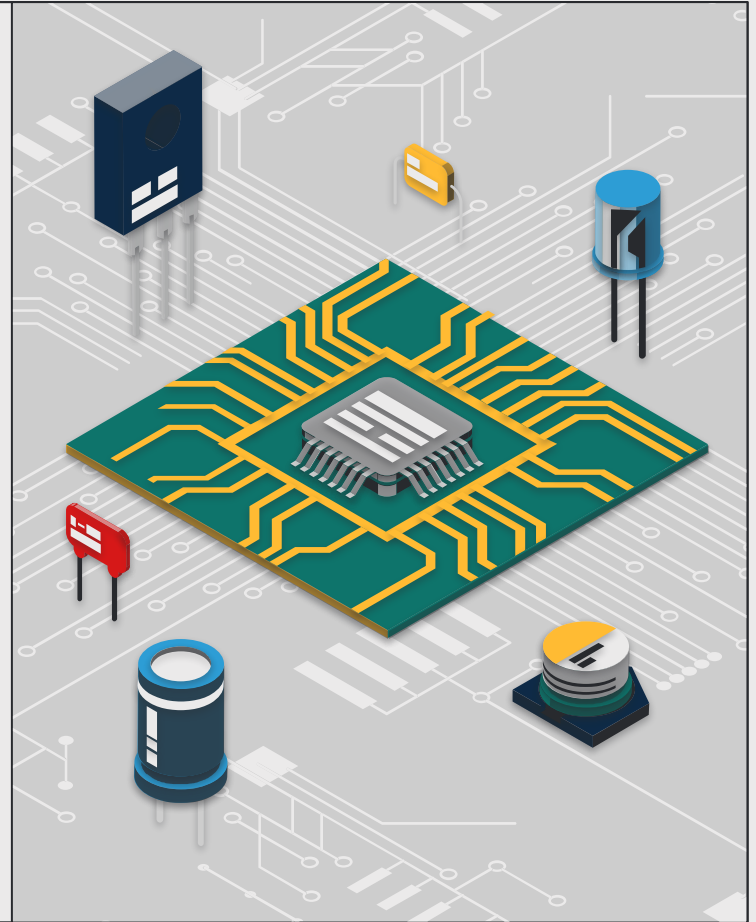


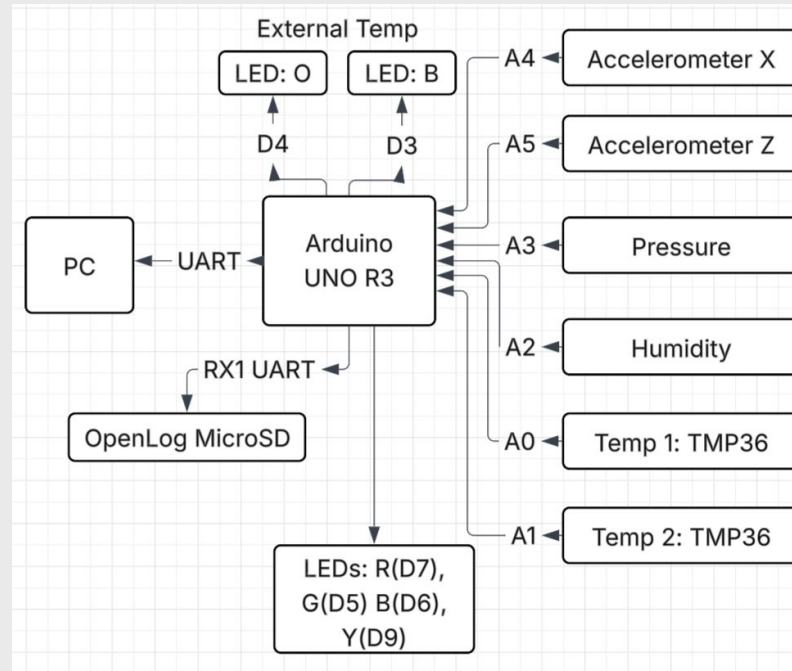
NASA Colorado Space Grant Consortium

Arduino Shield - Aidan Donnellan



Current Architecture

Block diagram for current Arduino UNO R3 Hat used for NASA CO. Space Grant Consortium



Current Component Selection

<u>Component</u>	<u>Purpose</u>	<u>Power Consumption (mA)</u>	<u>Weight (g)</u>
Arduino UNO R3	Brains	~20-75	~25
TMP36	Internal and External Temperature	<.5 (data sheet) → <1	~0.2 → 0.4
HIH-4030	Humidity Sensor	.2 (link)	~1
ADXL335	Accelerometer	.35 (data sheet)	~4.54
SSCDANN015 PD2A5	8-pin Pressure Sensor	<10, avg. 3 (datasheet)	~1
LED's	STD. 5mm (O, (2)B, R, G, Y)	16-18 (link)	~0.3
OpenLog MicroSD	Serial Logger	2-3 (idle), 10-20 (powered) (link)	3
PCB	-	-	~15
Header Pins	-	-	~1.8
<u>Total Power Consumption:</u> 119.35 (max)		<u>Total Weight:</u> ~52.04 (not considering battery)	

Feasibility Assessment – 1

<u>Component</u>	<u>Feasibility / Alternative</u>
Arduino UNO R3	Is bulky and heavy, alternatives such as the Nano Every or Teensy 4.x provide lightweight solutions with same, if not better, logic.
HIH-4030	Reliable, however, condensation or moisture exposure can impact performance.
SSCDANN015PD2A5	High quality, expensive pressure sensor. Good for stratospheric pressure accuracy
ADXL335	Low power, susceptible to signal noise
OpenLog	Necessary for data collection, other open log models exist if a specific device is not available.
TMP36	Good for sub-stratosphere testing, not suited for stratosphere temperatures.

Feasibility Assessment – 2

Component Feasibility Assessment: The primary feasibility issues lies not within components being unavailable, rather the thermal and power limits of certain components. At stratospheric temperatures, certain components such as the TMP36 are only rated down to -40 degrees celsius. Once the balloon passes into the stratosphere, the sensor is highly likely to flatline or give incorrect data, making this component in specific unsuitable for use.

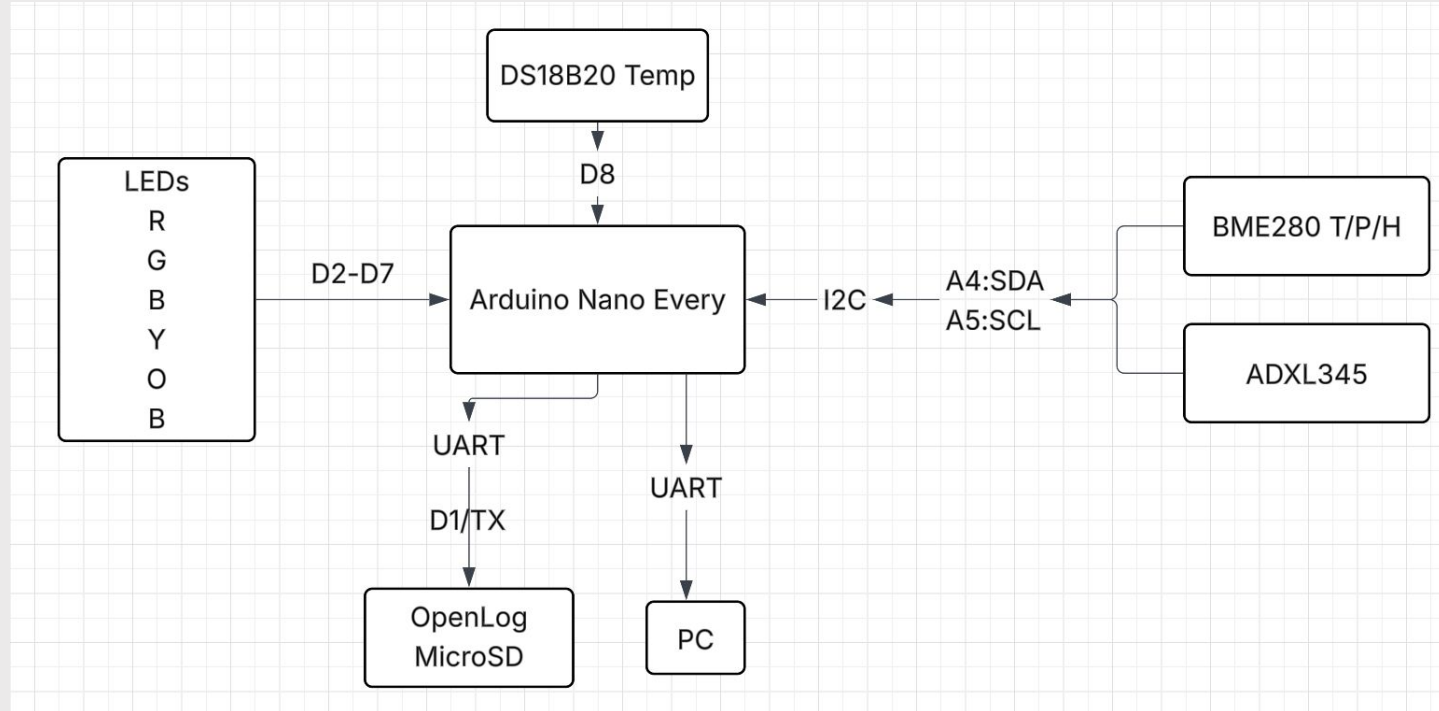
Component Unavailability: Most components used within the project have strong alternatives with similar ratings. However, weight and cost play a substantial role in component selection.

Overall: The current system is well suited for low-altitude testing; however, for NASA compliance and full scale launch, certain components should be reconsidered.

Pre-Phase 2 Plan: Design a PCB utilizing an Arduino Nano Every and BME280 temperature/humidity/barometric pressure sensor for decreased payload and form factor.



Revised Architecture



Component Selection

<u>Component</u>	<u>Purpose</u>	<u>Power Consumption (mA)</u>	<u>Weight (g)</u>
Arduino Nano Every	Brains	~18	5
DS18B20	External Temp	1	1
BME280	Internal Temp, Pressure, Humidity	0.1	15
ADXL345	Accelerometer	0.1	2
LED's	STD. 5mm (O, (2)B, R, G, Y)	16-18 (link)	2
OpenLog MicroSD	Serial Logger	2-3 (idle), 10-20 (powered) (link)	2
PCB	-	-	20
Header Pins	-	-	
<u>Total Power Consumption:</u> 135 (max), ~36 (status)		<u>Total Weight:</u> ~47 (not considering battery)	

BOM

Total Cost: \$104.16 (Single Board) → \$133.14 (3 Board) (pre-tax + shipping)

<u>Part</u>	<u>Cost (USD)</u> (pre-tax + shipping)	
Arduino Nano Every	12.90	
SparkFun ADXL345	20.95	
SparkFun BME280	16.95	
SparkFun OpenLog	17.50	
SparkFun DS18B20	10.95	
Resistors (x6)	0.04 → 0.24	
LED's (x6)	0.53 → 3.18	
Headers and Sockets	~5	
Screw Sockets	~2	
PCB (OSH Park)	8.69 sq. in. x \$5 = 43.47 (3 Boards) ~14.49	

Component Specifications

Arduino Nano Every

ATMega4809

- AVR CPU up to 20MHz
- 48 kB Flash
- 6 kB SRAM
- 256B EEPROM

ATSAMD11D14A

- Arm Cortex M0+ at up to 48MHz
- 16KB Flash
- 4KB SRAM
- 12Mbps USB 2.0 Interface

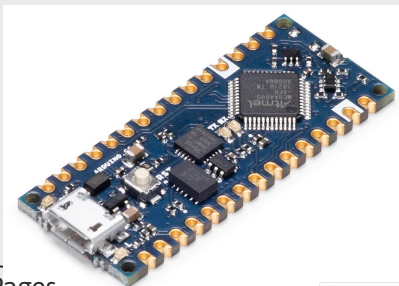
▪ **Peripherals**

- 1x 16-bit Timer/Counter with a dedicated period register and 3x compare channels
- 4x 16-bit Timer/Counter with input capture
- 1x 16-bit Real-Time Counter (RTC) running from an external crystal or an internal RC oscillator
- 4x USART with fractional baud rate generator, auto-baud, and start-of-frame detection
- 1x Master/slave Serial Peripheral Interface (SPI)
- 1x Dual mode Master/Slave TWI with dual address match 6x 16 bit Timers (1 dedicated to RTC)
- Event System for CPU independent and predictable inter-peripheral signaling
- Configurable Custom Logic (CCL) with up to four programmable Look-up Tables (LUT)
- 1x Analog Comparator (AC) with a scalable reference input
- Watchdog Timer with Window mode, with separate on-chip oscillator
- External interrupt on all general purpose pins

Why Arduino Nano Every

Arduino Nano Every

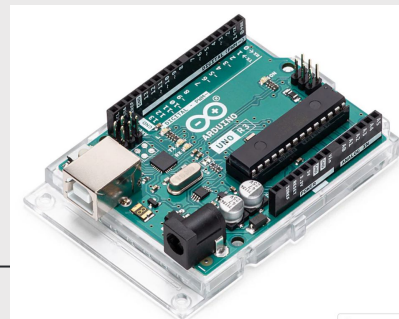
- Smaller Form Factor
- ATmega4809
 - 48 KB Flash
 - 6 KB SRAM
 - 256 Bytes EEPROM
 - CPU Speed: 20 MHz
 - Hardware UARTS: 4
 - ADC Channels: 16 (10-bit)



Photos from Product Pages

Arduino UNO

- Larger Form Factor
- ATmega328P
 - 32 KB Flash
 - 2KB SRAM
 - 1KB EEPROMs
 - CPU Speed: 20 MHz
 - Hardware UARTS: 1
 - ADC Channels: 8 (10-bit)



Component Specifications

SparkFun BME280

- Operating Voltage: 1.71V-3.6V
- Temp Range: -40 C to +85 C
- Humidity: 0% to 100%
- Pressure 300 hPa to 1100 hPa
- Altitude: 0 to 30,000 ft.
- Interfacing: I2C (Address 0x77) up to 3.4MHz



SparkFun ADXL345

- Operating Voltage: 2.0V - 3.6V
- Resolution: up to 13-bit
- Output Rate: 0.1 to 3200 Hz
- Interfacing: I2C (Address 0x53) and SPI
- Operating Temp: -40 C to + 85 C



SparkFun OpenLog

- Operating Voltage: 3.3V to 5V
- Baud: 300 to 115,200 bps
- Storage: microSD up to 32GB
- Processor: ATmega382P

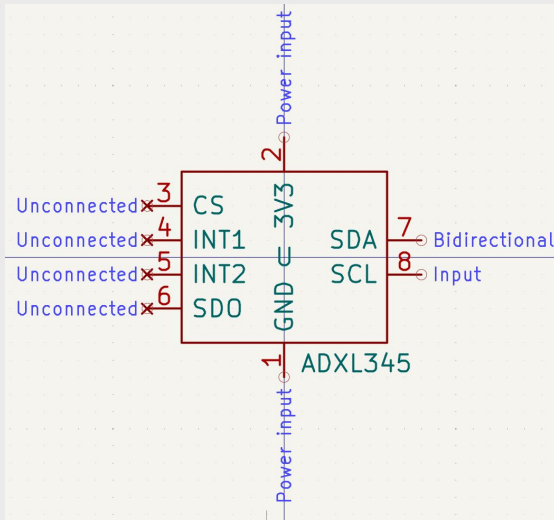


SparkFun DS18B20 Temp

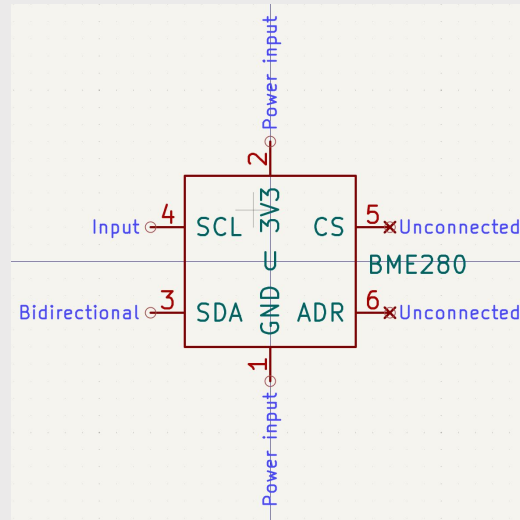
- Operating Voltage: 3.0V to 5.5V
- Operating Temp: -55 C to +125 C
- Resolution: 9 to 12 bits
- Communication: 1 wire interface
- 1m cable length

Custom Symbols

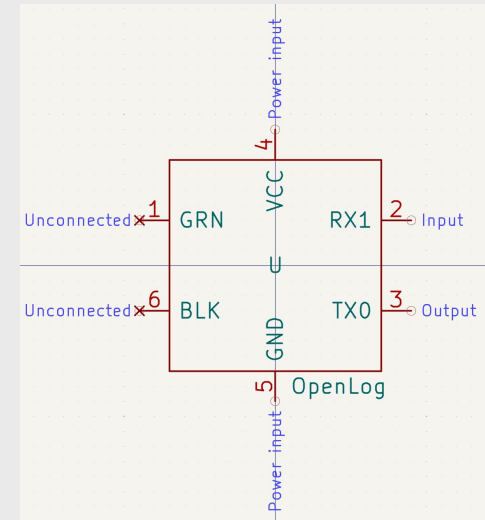
Creating Custom Symbols Using Symbol Editor



ADXL345 Module



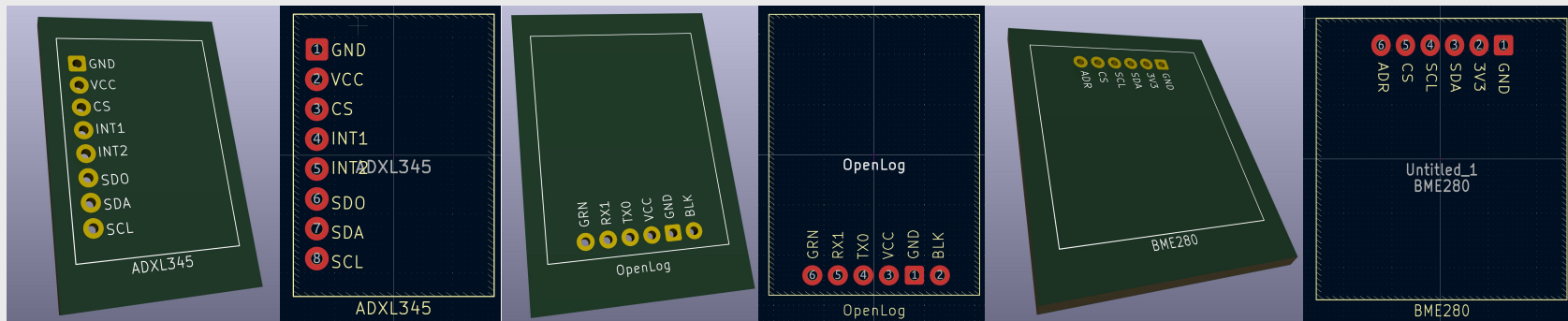
BME280 Module



OpenLog Module

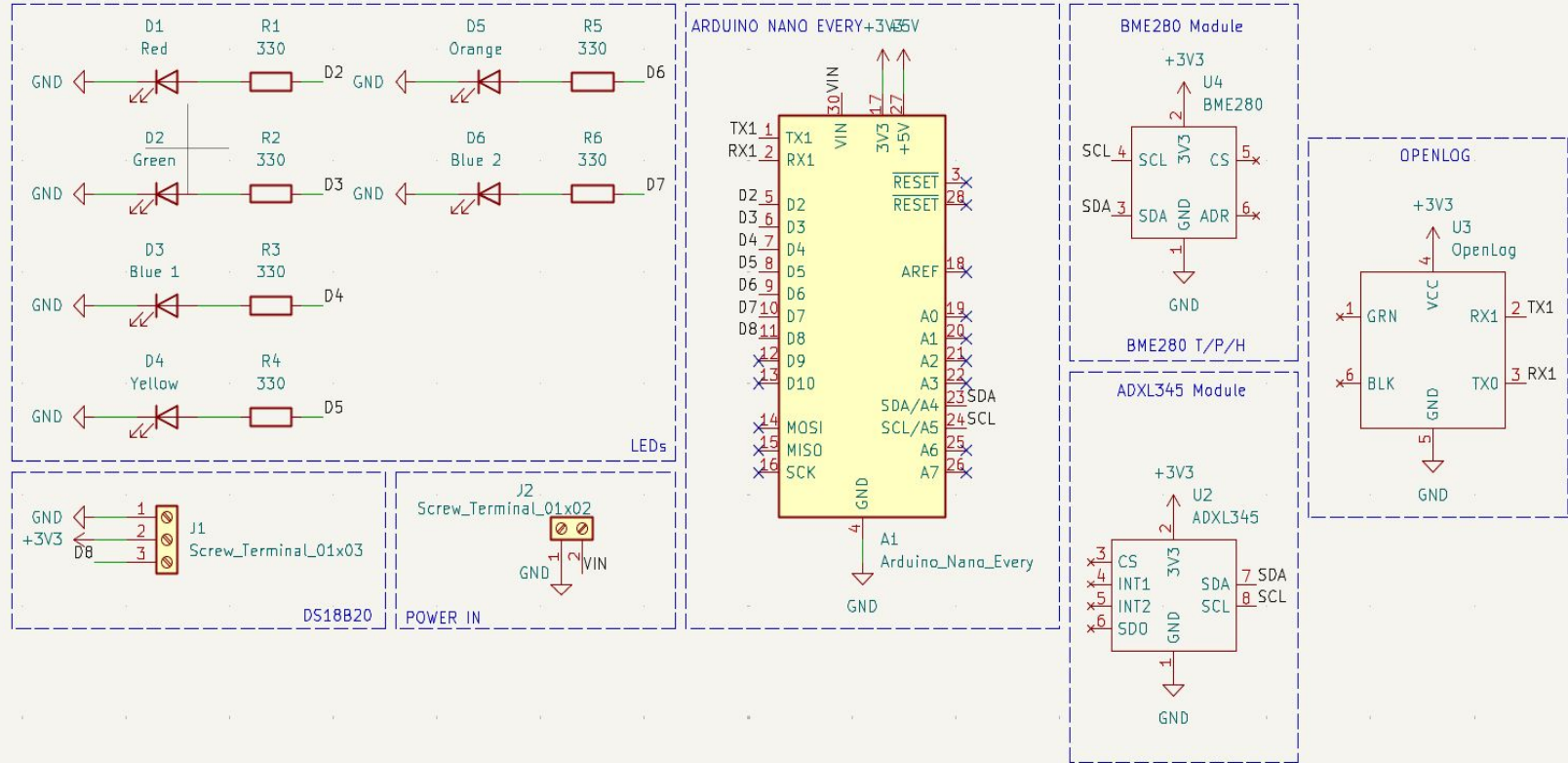
Custom Footprints

Creating Custom Footprints Using Footprint Editor and Footprint Assignment

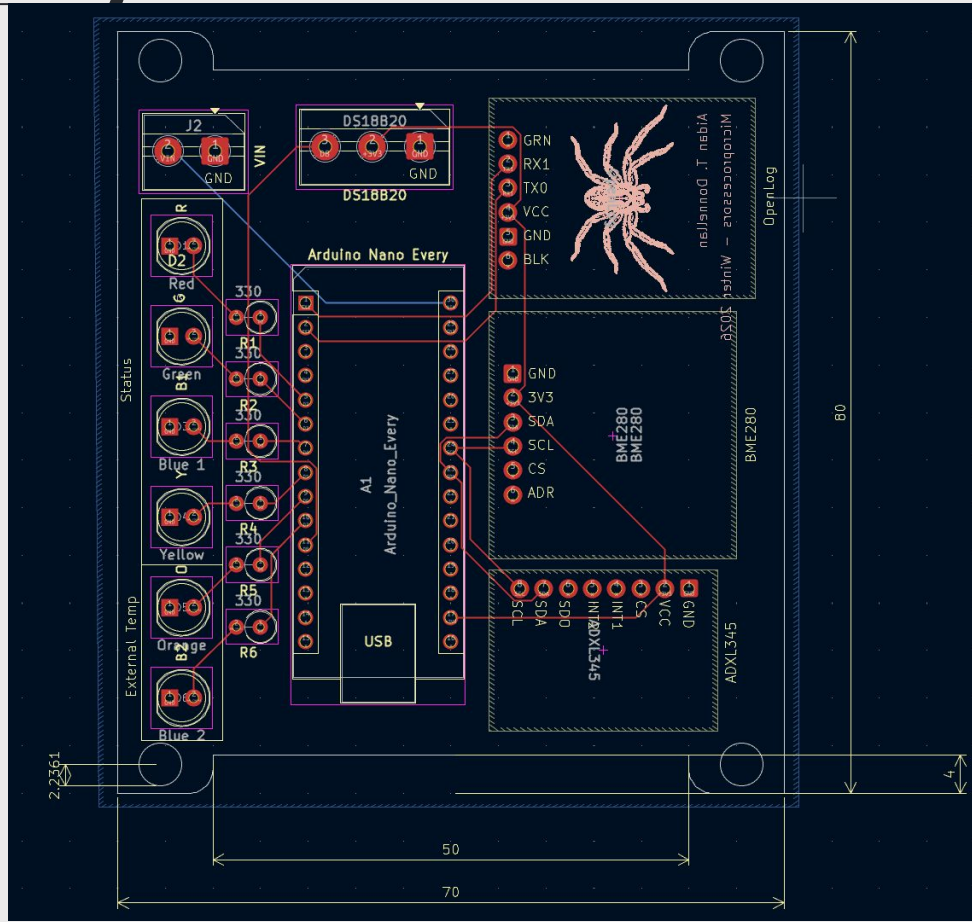


```
1 A1 - Arduino_Nano_Every : Module:Arduino Nano
2 D1 - Red : LED_THT:LED_D5.0mm
3 D2 - Green : LED_THT:LED_D5.0mm
4 D3 - Blue 1 : LED_THT:LED_D5.0mm
5 D4 - Yellow : LED_THT:LED_D5.0mm
6 D5 - Orange : LED_THT:LED_D5.0mm
7 D6 - Blue 2 : LED_THT:LED_D5.0mm
8 J1 - Screw_Terminal_01x03 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-03P_1x03_P5.00mm
9 R1 - 330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
10 R2 - 330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
11 R3 - 330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
12 R4 - 330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
13 R5 - 330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
14 R6 - 330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
15 U1 - BME280 : BME280:BME280_fp
16 U2 - ADXL345 : ADXL345:ADXL_fp
17 U3 - OpenLog : OpenLog:OpenLog_fp
```

Schematic



PCB Assembly



The image displays three views of a custom PCB design:

- Component Render:** A 3D perspective view showing the physical components placed on the PCB, including an Arduino Nano Every, various sensors (BME280, ADXL345, DS18B20), and a USB connector.
- Front:** A top-down view of the PCB showing the layout of components, including the Arduino Nano Every, BME280, ADXL345, DS18B20, and various connectors (USB, Status, External Temp).
- Back:** A bottom-up view of the PCB showing the layout of components, including the Arduino Nano Every, BME280, ADXL345, DS18B20, and various connectors (USB, Status, External Temp).

Component Render

Front

[Back](#)

Current Evaluation for Future Dev

Adding a 2x15 pin socket for extra modularity.

Re-evaluation of certain component choices that better suit stratospheric temperatures.



Documentation

KiCAD

- Arduino Nano Every 3D STP: [Arduino Nano Every 3D STP](#)
- SparkFun ADXL345 3D STP: [ADXL345 3D STP](#)
- BME280 Module: Lauren Ooghe
- SparkFun OpenLog: Lauren Ooghe
- HTM Workshop Playlist: [Playlist](#)

Reference Material

- Nasa Space Grant Consortium: [High Altitude Balloon](#)

Link to all files can be found here: [Donnellan Git](#)

Data Sheets

- Arduino Nano Every Data Sheet: [ADX00028](#)
- Arduino UNO R3 Data Sheet: [AD000066](#)
- SparkFun ADXL345 Data Sheet: [ADXL345](#)
- SparkFun BME280 Data Sheet: [BME280](#)
- SparkFun OpenLog Product Page: [OpenLog](#)
- DS18B20 Temp Data Sheet: [DS18B20](#)
- TMP36 Data Sheet: [TMP36](#)
- SparkFun HIH 4030 Product Page: [HIH 4030](#)
- SparkFun ADXL335 Data Sheet: [ADXL335](#)
- Honeywell 8-pin Pressure Sensor: [Pressure Sensor](#)